NNAMDI AZIKIWE UNIVERSITY, AWKA

BIO102 COURSE

TOPIC: ENVIRONMENTAL HEALTH AND DISEASES

KINGNAS

FOR

100 LEVEL STUDENTS

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ENVIRONMENTAL HEALTH AND DISEASES

Definition of Terms

Environment

Environment is defined as the complex of physical, chemical and biotic factors that act upon an organism or an ecological community and ultimately determines its form and survival. It is categorized into air environment, water environment, socio-economic environment and cultural environment.

Health

Health is a state of complete physical, mental, and social well being and not merely absence of disease or infirmity.

Environmental Health

Environmental health involves those aspects of the human health and diseases that are determined by the factors in the environment .It includes both the direct pathological effects of chemicals, radiation and some biological agents, and the effects on health and well-being of the broad physical, psychological, social and cultural environment.

Environmental health is also defined as the control of all factors in man's environment which exercise, or may exercise a deleterious effect on his physical development, health or survival. It comprised the aspect of human health, including quality of life, which is determined by physical, biological, chemical social and psychological factors in the environment.

It is an organized community efforts which monitor the relationship between man and his environment in other to achieve a better healthy living. The components of environmental health as specified by world health organization include: waste management, food hygiene and safety, pest and vector control, housing sanitation, air quality monitoring, occupational health and safety, water resources management and sanitation, environmental health impact and assessment, health education and promotion, collection and analysis of environmental data, epidemiological study and control of communicable diseases, noise pollution monitoring and prevention, control of frontiers , air and seaport and border crossing, protection of public amenities and recreational environment, promotion and enforcement of environmental health quality standards.

Causes of environmental health problems

The failure of the numerous efforts to address the problem of environmental health has been attributed to various factors. Prominent among these are unhealthy socio-cultural practices, poor environmental sanitation education and awareness, low literacy level, bad governance over the years, disregard for the rule of law and other forms of indiscipline.

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Environmental Diseases

Environmental diseases are illnesses and conditions that result from man-made environmental problems. Illnesses and conditions caused by factors in the environment are collectively called environmental diseases. Pesticides, chemicals, radiation, air pollution, and water pollution, are some of the man-made hazards that are believed to contribute to human illnesses. Potential illness-causing agents are everywhere: at home, at work, and at play. However, the likelihood of an individual developing a specific disease depends on the hazards present in their particular environment and their genetic susceptibility to a specific hazard. For example, x-ray technicians are at risk for radiation-induced illnesses, whereas coal miners are prone to lung diseases caused by inhalation of dust. Proper use of safeguards can prevent these and other environmental diseases.

What Are Some of the Common Environmental Diseases?

1. Lung Diseases

Any substance other than air that is breathed into the lungs has the potential to cause damage to these organs. For example, air pollution, including smoke from other peoples' cigarettes (secondhand smoke), and workplace chemicals can lead to lung diseases. Examples of lung diseases include:

Asthma, a condition in which breathing is difficult, affects millions of Americans. Environmental triggers for asthma are everywhere and include naturally occurring triggers such as animal dander, plant pollen, dust, and mold, and manmade triggers such as chemicals. Not everyone is sensitive to these triggers, but many people are sensitive to some of them.

Black lung disease is an illness in which coal miners' lungs become coated with coal dust, causing a chronic condition in which breathing becomes difficult and painful.

Bronchitis, an inflammation of the airways of the lungs, can be caused by breathing in certain chemicals or smoke. Welders and fire fighters are some of the people at risk for this condition. Smokers are also at increased risk for the development of bronchitis and lung cancer.

Breathing asbestos (a natural mineral fiber) can lead to asbestosis (as-bes-TO-sis), a severe lung disorder, and lung cancer. Schools, homes, and businesses that have asbestos in them as a fire retardant put people at risk when the asbestos dust begins to leak into the air during repairs and renovations. These buildings used asbestos in the walls and ceilings as insulation before laws were passed to ban their use.

Silicosis is a lung disease caused by exposure to the silica dust in clay. Pottery workers are at risk of developing this disease.

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2. Cancers

In addition to lung cancer, other cancers have been linked to environmental toxins (poisons). For example, pesticides, herbicides, and radioactive substances have the potential to cause cancer. There are almost 2,000 chemicals that are suspected of causing cancer. Of these, only several hundred have had their use restricted by law in the United States.

Asbestos, chromium, and coal tar have been linked to lung cancer. Construction workers, welders, and steelworkers may be repeatedly exposed to these compounds. People working in plastics manufacturing are at risk for liver or bladder cancer. People who work with radioactive substances are at increased risk for cancer caused by radiation. Fortunately, legal restrictions and careful oversight of hazardous materials can reduce risks.

3. Gulf War Syndrome

Many United States veterans have complained about disabling symptoms that they attribute to their participation in the 1991 Persian Gulf War. Gulf War Syndrome encompasses symptoms such as chronic fatigue, aching muscles and joints, skin rashes, memory loss, miscarriages, and babies born with birth defects.

Gulf War Syndrome is a mysterious condition. While veterans are experiencing very real medical problems, physicians and scientists cannot agree on the source of these problems. Some people believe that the syndrome actually consists of multiple illnesses for which symptoms overlap. Veterans could be reacting to chemical weapons, biological weapons, pesticides, vaccines, oil fires, or infectious diseases that they were exposed to in the Gulf War. A presidential commission set up to study the controversy surrounding the Gulf War Syndrome proposed that stress was the major cause behind the symptoms the veterans were experiencing. A decade after it began, the controversy continues as conflicting reports emerge.

4. Birth Defects

Infertility, miscarriage, stillbirth (the baby is born dead), childhood cancer, and birth defects may have links to various environmental toxins. When a pregnant woman is exposed to lead, her child has a higher than usual risk of being born with behavior and nervous system problems. Exposure to radiation, chemical wastes, pesticides, solvents, paints, lead, and methyl mercury all can cause problems in a developing fetus.

5. Chemical Poisonings

Lead is a serious environmental hazard to children in many parts of the world, including the United States. It affects children's mental and physical development, and high doses can cause paralysis and death. People can be exposed to lead through lead paint, leaded gasoline, lead water pipes, and certain ceramics. Although lead is no longer used in most of these products in the United States, it can still be found in older homes and in some imported products.

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The metals mercury and cadmium can cause nerve damage, cancer, and liver and skin diseases. Mercury has been used since the beginning of the nineteenth-century in many industrial processes. In earlier times, people who worked with mercury were often unknowingly poisoned. Mercury can accumulate in the food chain and present a health risk. For example, some fish in the Great Lakes are contaminated with mercury they acquired through eating plants and other fish. Eating a lot of these contaminated fish can transfer unhealthy levels of mercury to a person. The effects are cumulative, (build up), because the body cannot rid itself of mercury. Mercury can be found in fluorescent lights, latex paint, batteries, dental fillings, and mercury thermometers.

Other sources of environmental poisoning can arise from the manufacture of refrigerants, plastics, and other industrial products and from the manufacture and misuse of pesticides.

6. **Neglected tropical diseases (NTDs),**

Neglected tropical diseases (NTDs), a group of communicable diseases found in predominantly tropical and subtropical conditions, affect the poorest populations who often live in remote, rural areas, in urban slums, or in conflict zones. The Centre for Disease Control (CDC) reports that one hundred percent of low-income countries are concurrently affected by at least five neglected tropical diseases. With little political voice, neglected tropical diseases typically have low status among public health priorities. In 2012, the WHO published a comprehensive plan that established targets for the reduction of NTDs, representing a shift in these priorities. In 2010, an estimated 2 billion individuals needed treatment for neglected tropical diseases. By 2016, the estimate decreased to 1.5 billion confirming the great gains have been through approaches in treatment and prevention.

Treatment and Prevention

Many measures can be taken almost immediately to reduce the environmental disease burden, such as the promotion of safe household water storage, improved hygiene, and the control of the use of toxic substances in the home and workplace. Though many of these diseases can be treated with simple, inexpensive medications, global strategies to control NTDs also include **preventive chemotherapy, intensified case management, vector control, veterinary public health measures and improved water quality and sanitation.**

In 2013, the WHO categorized seventeen diseases as neglected tropical diseases, namely **buruli ulcer, Chagas diseases, dengue fever, guinea-worm, liver flukes, trachoma, tapeworms, yaws, soil-transmitted helminthiases, river blindness, rabies, lymphatic filariasis, leishmaniasis, schistosomiasis, sleeping sickness and hydatid disease. Recent additions include mycetoma and snakebite envenoming.** This text provides a comprehensive review of a few of the major neglected tropical diseases:

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Trachoma

Trachoma is the leading cause of preventable blindness of infectious origins caused by the bacterium *Chlamydia trachomatis*. Trachoma is spread easily by contact with an infected person's hands or clothing, or by flies that come in contact with the eyes or nose of an infected person. Infected individuals do not instantly go blind; instead, the disease manifests gradually. Scarring from repeated infections causes the eyelashes to turn inward and scratch the cornea, a condition known as trichiasis, leading slowly and painfully to complete blindness. Since trachoma is transmitted through close personal contact, it tends to occur in clusters, often infecting entire families and communities. Estimates suggest that 1.9 million people are blind due to trachoma while another 21 million are actively infected with the disease.

Blindness from trachoma strikes adults in their prime years, hindering their ability to care for themselves and their families. Women, traditionally the caretakers of the home, are four times more likely than men to be affected by trichiasis due to increased contact with infected children and limited financial agency in health-related decisions necessary for treatment and prevention.

Treatment and Prevention

Trachoma is treatable and preventable with an approach known as the SAFE strategy. Recommended by the World Health Organization, the SAFE strategy is a comprehensive public health approach that combines treatment (Surgery and Antibiotics) with prevention (Facial cleanliness and Environmental improvement). The strategy works as follows:

S= Surgery to correct advances of the disease
A= Antibiotics, such as azithromycin, to treat infected individuals
F= Face washing to reduce transmission
E= Environmental improvement through increased access to clean water and improved sanitation

Through the work of national governments and various partners, the implementation of the SAFE Strategy has led to measurable results in the elimination of trachoma. As of 2016, over 260,000 individuals received surgery for trichiasis and over 85 million antibiotic treatments were given worldwide.

River blindness

Onchocerciasis, also known as river blindness, infects approximately 18 million people, of which the majority live in sub-Saharan Africa. Over 120 million globally are at risk for contracting river blindness. Approximately 6.5 million individuals experience severe itching due to onchocerciasis and another 270,000 have been left permanently blind from the disease. People living near the fast-moving streams of Africa are at high risk since these areas are breeding grounds for the *Simulium* black fly. Infected flies carry larvae of the filarial parasitic worm *Onchocerca volvulus* from person to person. Larvae, which enter the skin at the black fly

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bite site, move throughout the body and then die, resulting in a variety of health problems including skin rashes and de-pigmentation, itching, and skin nodules. Visual impairment and blindness occur if larvae enter the eye.

Treatment and Prevention

The severe socioeconomic impact of river blindness prompted the creation of the highly successful Onchocerciasis Control Program in 1974. This program focused on 11 countries in West Africa that aimed to make ivermectin treatment more accessible and included intensified spraying of black fly breeding sites with environmentally safe insecticides. The program ended in 2002 after the successful end to transmission of the disease in 10 of the 11 participating countries. In 2016, over 133 million treatments were provided globally. Current elimination barriers include political uncertainty and competing health priorities, though progress has been made and four countries within Central and South America have successfully eliminated river blindness since 2013.

Guinea Worm disease

Guinea worm disease, formally known as Dracunculiasis, is caused by the parasitic worm *Dracunculus medinensis*. It is transmitted through ingestion of stagnant water that is contaminated with guinea worm larvae. In the human body, the larvae are released and migrate into body tissues, and develop into adult worms. Female worms move through the person's tissue, causing intense pain, and eventually emerge through the skin, producing oedema, a blister and eventually an ulcer, accompanied by fever. Though rarely fatal, symptoms include joint pain, vomiting and severe burning. There are no drugs available for treatment of the disease and the worm is instead physically removed over a period of several weeks. Prevention is key in the eradication efforts of guinea worm, specifically by protecting water sources and filtering contaminated water.

Treatment and Prevention

The global transmission of guinea worm disease has significantly been reduced with only three reported cases as of April 2018 within the endemic countries of Chad, Ethiopia and Mali. The fight to end guinea worm was greatly influenced by The Carter Center which led the Guinea Worm Eradication Program along with partnerships with local ministries of health and global organizations beginning in the 1980s. The program's success was a direct result of community-based interventions that focused on water filtration and health education. Guinea worm disease is on track to be the first eradicated neglected tropical diseases due to these concerted efforts on a global and local level.

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Schistosomiasis

Schistosomiasis is acquired through contact with parasitic worms that are released by freshwater snails. Â Inside the body, the larvae develop into worms. Female worms release thousands of eggs which are passed out of the body and released into bodies of freshwater by urination or defecation. They migrate to snails and begin the cycle again. Some eggs trapped in the body migrate to specific organs and inflict major damage such as scarring and tearing of the bladder and kidneys. Estimates suggest that over 205 million individuals required treatment for schistosomiasis in 2015. Schistosomiasis is prevalent in 78 countries, with 92 % of those requiring treatment living in sub-Saharan Africa. The most susceptible populations include children under age 14, as well as individuals with responsibilities and occupations near infested water sources.

Treatment and Prevention

Improved sanitation and hygiene are critical to the elimination of schistosomiasis. Additional measures include efforts to eliminate the snail species required for transmission. Preventative chemotherapy, where individuals are treated without being diagnosed, has been shown to be successful in decreasing the disease burden of schistosomiasis globally. Praziquantel is the only recommended drug to treat schistosomiasis and has also been shown to reduce the severity of symptoms in those re-infected with the disease.

Soil Transmitted Helminth Infections

Hookworm, roundworm and whipworm make up the group of intestinal soil-transmitted infections and are among the most common infections in the world. These infections are associated with poverty and poor sanitation given their transmission through contaminated food, soil, and feces. There are an estimated 1.5 billion individuals infected by soil-transmitted helminths.

Ascariasis, also known as roundworm, is an intestinal infection caused by the soil transmitted parasitic worm Ascarislumbricoidesart. It is most prevalent in warm tropical and sub-tropical climates in Sub-Saharan Africa and Southeast Asia. The disease flourishes in areas with poor sanitation and poorly irrigated crops treated with wastewater. It is the most common human worm infection. Between 807 million to 1.2 billion people worldwide are infected with ascariasis. It is transmitted by accidental ingestion of ascaris eggs in contaminated food, water or soil. The larvae migrate into the bloodstream and cause severe coughing and wheezing.

Whipworms, caused by the soil transmitted parasitic worm *Trichuris trichiura* affect approximately 604-795 million individuals. Individuals infected by only a few worms usually do not experience symptoms while those with a higher parasitic load may experience weakness, anemia, rectal prolapse and even hinder growth in children.

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Hookworm, caused by the soil transmitted parasitic worms *Necator americanus* and *Ancylostoma duodenale* affect approximately 576-740 million individuals globally. Infected individuals typically experience gastrointestinal symptoms.

Prevention & Treatment

Albendazole and mebendazole are the recommended medications for intestinal parasitic worms. These medications are easily accessible through the ministries of health in endemic countries. The WHO adopted a resolution to eliminate the disease burden due to soil-transmitted helminth infections by 2020 which would require periodic deworming of 75% of all at-risk school children by 2020. The program has been able to achieve some success in reducing the severity of worm infections in some areas, particularly in Africa. In 2017, 485 million school age children were treated with deworming tablets. Education efforts aimed at prevention through proper sanitation, hand washing, and food preparation techniques are critical to reducing the incidence of the disease in regions of Africa.

Lymphatic Filariasis

Lymphatic Filariasis, also known as elephantiasis is common in tropical and sub-tropical climates, in both rural and urban areas where infected mosquitoes tend to breed. Over 850 million people are at risk for becoming infected with lymphatic filariasis with approximately 40 million currently affected by the disease. Lymphatic Filariasis is transmitted when infected mosquitoes bite humans and pass microfilariae into the skin. Larvae migrate through the body and settle in the lymphatic system, where they lay eggs that pass into the blood. The worms can live for up to six years here. When they die, they can cause severe disfiguring ailments. Initially, no symptoms may be present, but as the disease progresses, the following may occur: swelling or hardening of the skin, severe swelling/lymphedema of extremities or genitals, and "filarial fevers" which are episodes of severe swelling in chronic patients. The disease has been associated with social stigma leading to emotional and mental suffering.

Treatment and Prevention

Unfortunately, there is no cure or vaccine for the disease, but symptoms can be managed with a treatment plan. The treatment to reduce the symptoms of Lymphatic Filariasis includes a combination of albendazole, diethylcarbamazine (DEC), and ivermectin medicines. Evidence suggests that treatment with these medications concurrently can eliminate the microfilaria from the bloodstream in a few weeks times. Careful cleansing of the skin to remove bacteria can reduce or reverse skin or tissue damage. The 1997 WHO's Global Program to Eliminate Lymphatic Filariasis was implemented to stop the spread of transmission and to lessen the severity of the disease in those already infected. Hygiene education efforts aimed at reducing acute episodes of filarial fevers have been effective at lessening the social stigmas associated with the manifestations of the disease. Since 1997, 18 countries have begun the process to

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confirm elimination while mass treatment is currently ongoing in 45 countries demonstrating great strides in the elimination of lymphatic filariasis.

Sick Building Syndrome

Sick Building Syndrome describes an elusive health problem in which people attribute a variety of symptoms to the buildings where they work. Common complaints include headaches, dizziness, nausea, tiredness, concentration problems, sensitivity to odors, dry itchy skin, a dry cough, and irritated eyes, nose, and throat. Generally, as soon as affected people leave the building, their symptoms vanish and they feel well again.

Sick Building Syndrome is elusive in that neither one specific illness nor a common cause has been identified. No specific set of symptoms is common to all people complaining of Sick Building Syndrome. While nothing has been proven, theories as to the causes of Sick Building Syndrome abound. Many factors may contribute to it, including humidity, poor ventilation, and poor temperature control. Pollution from outdoor sources, (for example car exhaust, pollen, or smoke) or chemicals from inside a building, (cleaning supplies, glues, upholstery, or copy machine chemicals), may affect people who are sensitive to them. Bacteria, viruses, and molds also can invade buildings and make people sick.

ANTHROPOGENIC IMPACTS OF THE ENVIRONMENT

Human impact on the environment or anthropogenic impact on the environment includes changes to biophysical environments and ecosystems, biodiversity, and natural resources caused directly or indirectly by humans, including global warming environmental degradation (such as ocean acidification), mass extinction and biodiversity loss, ecological crisis, and ecological collapse. Modifying the environment to fit the needs of society is causing severe effects, which become worse as the problem of human overpopulation continues. Some human activities that cause damage (either directly or indirectly) to the environment on a global scale include human reproduction, overconsumption, overexploitation, pollution, and deforestation, to name but a few. Some of the problems, including global warming and biodiversity loss pose an existential risk to the human race, and overpopulation causes those problems.

The term anthropogenic designates an effect or object resulting from human activity. The term was first used in the technical sense by Russian geologist Alexey Pavlov, and it was first used in English by British ecologist Arthur Tansley in reference to human influences on climax plant communities. The atmospheric scientist Paul Crutzen introduced the term "Anthropocene" in the mid-1970s. The term is sometimes used in the context of pollution emissions that are produced from human activity but also applies broadly to all major human impacts on the environment.

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Human overpopulation and effects on the environment

David Attenborough described the level of human population on the planet as a multiplier of all other environmental problems. In 2013, he described humanity as "a plague on the Earth" that needs to be controlled by limiting population growth.

Some deep ecologists, such as the radical thinker and polemicist PenttiLinkola, see human overpopulation as a threat to the entire biosphere. In 2017, over 15,000 scientists around the world issued a second warning to humanity which asserted that rapid human population growth is the "primary driver behind many ecological and even societal threats."

Overconsumption

Overconsumption is a situation where resource use has outpaced the sustainable capacity of the ecosystem. It can be measured by the ecological footprint, a resource accounting approach which compares human demand on ecosystems with the amount of planet matter ecosystems can renew. Estimates indicate that humanity's current demand is 70% higher than the regeneration rate of all of the planet's ecosystems combined. A prolonged pattern of overconsumption leads to environmental degradation and the eventual loss of resource bases.

Humanity's overall impact on the planet is affected by many factors, not just the raw number of people. Their lifestyle (including overall affluence and resource utilization) and the pollution they generate (including carbon footprint) are equally important. In 2008, The New York Times stated that the inhabitants of the developed nations of the world consume resources like oil and metals at a rate almost 32 times greater than those of the developing world, who make up the majority of the human population.

Technology

The applications of technology often result in unavoidable and unexpected environmental impacts, which according to the I = PAT equation is measured as resource use or pollution generated per unit GDP. Environmental impacts caused by the application of technology are often perceived as unavoidable for several reasons. First, given that the purpose of many technologies is to exploit, control, or otherwise "improve" upon nature for the perceived benefit of humanity while at the same time the myriad of processes in nature have been optimized and are continually adjusted by evolution, any disturbance of these natural processes by technology is likely to result in negative environmental consequences. Second, the conservation of mass principle and the first law of thermodynamics (i.e., conservation of energy) dictate that whenever material resources or energy are moved around or manipulated by technology, environmental consequences are inescapable. Third, according to the second law of thermodynamics, order can be increased within a system (such as the human economy) only by increasing disorder or entropy outside the system (i.e., the environment). Thus, technologies can create "order" in the

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human economy (i.e., order as manifested in buildings, factories, transportation networks, communication systems, etc.) only at the expense of increasing "disorder" in the environment.

Environmental impact of agriculture

The environmental impact of agriculture varies based on the wide variety of agricultural practices employed around the world. Ultimately, the environmental impact depends on the production practices of the system used by farmers. The connection between emissions into the environment and the farming system is indirect, as it also depends on other climate variables such as rainfall and temperature.

The environmental impact of agriculture involves a variety of factors from the soil, to water, the air, animal and soil diversity, plants, and the food itself. Some of the environmental issues that are related to agriculture are climate change, deforestation, genetic engineering, irrigation problems, pollutants, soil degradation, and waste.

The environmental impact of fishing can be divided into issues that involve the availability of fish to be caught, such as overfishing, sustainable fisheries, and fisheries management; and issues that involve the impact of fishing on other elements of the environment, such as by-catch and destruction of habitat such as coral reefs.

Environmental impact of irrigation

The environmental impact of irrigation includes the changes in quantity and quality of soil and water as a result of irrigation and the ensuing effects on natural and social conditions at the tail-end and downstream of the irrigation scheme. The impacts stem from the changed hydrological conditions owing to the installation and operation of the scheme. An irrigation scheme often draws water from the river and distributes it over the irrigated area. As a hydrological result it is found that the:

- downstream river discharge is reduced
- evaporation in the scheme is increased
- groundwater recharge in the scheme is increased
- level of the water table rises
- drainage flow is increased.

These may be called direct effects.

Effects on soil and water quality are indirect and complex, and subsequent impacts on natural, ecological and socio-economic conditions are intricate. In some, but not all instances, water logging and soil salinization can result. However, irrigation can also be used, together with soil drainage, to overcome soil salinization by leaching excess salts from the vicinity of the root zone.

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Irrigation can also be done extracting groundwater by (tube) wells. As a hydrological result it is found that the level of the water descends. The effects may be water mining, land/soil subsidence, and, along the coast, saltwater intrusion.

Irrigation projects can have large benefits, but the negative side effects are often overlooked. Agricultural irrigation technologies such as high powered water pumps, dams, and pipelines are responsible for the large-scale depletion of fresh water resources such as aquifers, lakes, and rivers. As a result of this massive diversion of freshwater, lakes, rivers, and creeks are running dry, severely altering or stressing surrounding ecosystems, and contributing to the extinction of many aquatic species.

It increases the moisture contents of the soil and high humidity.

Environmental impact of coal mining and burning

The environmental impact of coal mining and -burning is diverse. Legislation passed by the US Congress in 1990 required the United States Environmental Protection Agency (EPA) to issue a plan to alleviate toxic air pollution from coal-fired power plants.

Environmental impact of electricity generation

The environmental impact of electricity generation is significant because modern society uses large amounts of electrical power. This power is normally generated at power plants that convert some other kind of energy into electricity. Each such system has advantages and disadvantages, but many of them pose environmental concerns.

Environmental impact of nuclear power

The environmental impact of nuclear power results from the nuclear fuel cycle processes including mining, processing, transporting and storing fuel and radioactive fuel waste. Released radioisotopes pose a health danger to human populations, animals and plants as radioactive particles enter organisms through various transmission routes.

Radiation is a carcinogen and causes numerous effects on living organisms and systems. The environmental impacts of nuclear power plant disasters such as the Chernobyl disaster, the Fukushima Daiichi nuclear disaster and the Three Mile Island accident, among others, persist indefinitely, though several other factors contributed to these events including improper management of failsafe systems and natural disasters putting uncommon stress on the generators. The radioactive decay rate of particles varies greatly, dependent upon the nuclear properties of a particular isotope. Radioactive Plutonium-244 has a half-life of 80.8 million years, which indicates the time duration required for half of a given sample to decay, though very little plutonium-244 is produced in the nuclear fuel cycle and lower half-life materials have lower activity thus giving off less dangerous radiation.

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Environmental impact of the oil shale industry

The environmental impact of the oil shale industry includes the consideration of issues such as land use, waste management, and water and air pollution caused by the extraction and processing of oil shale. Surface mining of oil shale deposits causes the usual environmental impacts of open-pit mining. In addition, the combustion and thermal processing generate waste material, which must be disposed of, and harmful atmospheric emissions, including carbon dioxide, a major greenhouse gas. Experimental in-situ conversion processes and carbon capture and storage technologies may reduce some of these concerns in future, but may raise others, such as the pollution of groundwater.

Environmental impact of petroleum

The environmental impact of petroleum is often negative because it is toxic to almost all forms of life. Petroleum, a common word for oil or natural gas, is closely linked to virtually all aspects of present society, especially for transportation and heating for both homes and for commercial activities.

Environmental impact of reservoirs

The environmental impact of reservoirs is coming under ever increasing scrutiny as the world demand for water and energy increases and the number and size of reservoirs increases.

Dams and the reservoirs can be used to supply drinking water, generate hydroelectric power, increasing the water supply for irrigation, provide recreational opportunities and flood control. However, adverse environmental and sociological impacts have also been identified during and after many reservoir constructions. Although the impact varies greatly between different dams and reservoirs, common criticisms include preventing sea-run fish from reaching their historical mating grounds, less access to water downstream, and a smaller catch for fishing communities in the area. Advances in technology have provided solutions to many negative impacts of dams but these advances are often not viewed as worth investing in if not required by law or under the threat of fines. Whether reservoir projects are ultimately beneficial or detrimental—to both the environment and surrounding human populations— has been debated since the 1960s and probably long before that. In 1960 the construction of LlynCelyn and the flooding of CapelCelyn provoked political uproar which continues to this day. More recently, the construction of Three Gorges Dam and other similar projects throughout Asia, Africa and Latin America have generated considerable environmental and political debate.

Manufactured products

Cleaning agents

Environmental impact of cleaning agents

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The environmental impact of cleaning agents is diverse. In recent years, measures have been taken to reduce these effects.

Environmental impact of nanotechnology

Nanotechnology's environmental impact can be split into two aspects: the potential for nanotechnological innovations to help improve the environment, and the possibly novel type of pollution that nanotechnological materials might cause if released into the environment. As nanotechnology is an emerging field, there is great debate regarding to what extent industrial and commercial use of nanomaterials will affect organisms and ecosystems.

Environmental impact of paint

The environmental impact of paint is diverse. Traditional painting materials and processes can have harmful effects on the environment, including those from the use of lead and other additives. Measures can be taken to reduce environmental impact, including accurately estimating paint quantities so that wastage is minimized, use of paints, coatings, painting accessories and techniques that are environmentally preferred. Some scientists suggest that by 2050 there could be more plastic than fish in the oceans.

Environmental impact of pesticides

The environmental impact of pesticides is often greater than what is intended by those who use them. Kills both target and non-target organisms. Over 98% of sprayed insecticides and 95% of herbicides reach a destination other than their target species, including non-target species, air, water, bottom sediments, and food. Pesticide contaminates land and water when it escapes from production sites and storage tanks, when it runs off from fields, when it is discarded, when it is sprayed aerially, and when it is sprayed into water to kill algae.

The amount of pesticide that migrates from the intended application area is influenced by the particular chemical's properties: its propensity for binding to soil, its vapor pressure, its water solubility, and its resistance to being broken down over time. Factors in the soil, such as its texture, its ability to retain water, and the amount of organic matter contained in it, also affect the amount of pesticide that will leave the area. Some pesticides contribute to global warming and the depletion of the ozone layer.

Can bring about resistance which warrants that a greater dosage is used.

Environmental impact of mining

The environmental impact of mining includes erosion, formation of sinkholes, loss of biodiversity, and contamination of soil, groundwater and surface water by chemicals from mining processes. In some cases, additional forest logging is done in the vicinity of mines to increase the available room for the storage of the created debris and soil. Besides creating

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environmental damage, the contamination resulting from leakage of chemicals also affect the health of the local population. Mining companies in some countries are required to follow environmental and rehabilitation codes, ensuring the area mined is returned to close to its original state. Some mining methods may have significant environmental and public health effects.

Environmental impact of transport

The environmental impact of transport is significant because it is a major user of energy, and burns most of the world's petroleum. This creates air pollution, including nitrous oxides and particulates, and is a significant contributor to global warming through emission of carbon dioxide, for which transport is the fastest-growing emission sector. By subsector, road transport is the largest contributor to global warming.

Environmental regulations in developed countries have reduced the individual vehicles emission; however, this has been offset by an increase in the number of vehicles, and more use of each vehicle. Some pathways to reduce the carbon emissions of road vehicles considerably have been studied. Energy use and emissions vary largely between modes, causing environmentalists to call for a transition from air and road to rail and human-powered transport, and increase transport electrification and energy efficiency.

Other environmental impacts of transport systems include traffic congestion and automobile-oriented urban sprawl, which can consume natural habitat and agricultural lands. By reducing transportation emissions globally, it is predicted that there will be significant positive effects on Earth's air quality, acid rain, smog and climate change.

The health impact of transport emissions is also of concern. A recent survey of the studies on the effect of traffic emissions on pregnancy outcomes has linked exposure to emissions to adverse effects on gestational duration and possibly also intrauterine growth.

Environmental impact of aviation

The environmental impact of aviation occurs because aircraft engines emit noise, particulates, and gases which contribute to climate change and global dimming. Despite emission reductions from automobiles and more fuel-efficient and less polluting turbofan and turboprop engines, the rapid growth of air travel in recent years contributes to an increase in total pollution attributable to aviation. In the EU, greenhouse gas emissions from aviation increased by 87% between 1990 and 2006. Among other factors leading to this phenomenon are the increasing number of hypermobile travellers and social factors that are making air travel commonplace, such as frequent flyer programs.

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Environmental impact of roads

The environmental impact of roads includes the local effects of highways (public roads) such as on noise, light pollution, water pollution, habitat destruction/disturbance and local air quality; and the wider effects including climate change from vehicle emissions. The design, construction and management of roads, parking and other related facilities as well as the design and regulation of vehicles can change the impacts to varying degrees.

Environmental impact of shipping

The environmental impact of shipping includes greenhouse gas emissions and oil pollution. In 2007, carbon dioxide emissions from shipping were estimated at 4 to 5% of the global total, and estimated by the International Maritime Organization (IMO) to rise by up to 72% by 2020 if no action is taken. There is also a potential for introducing invasive species into new areas through shipping, usually by attaching themselves to the ship's hull.

Environmental impact of bush burning

It reduces the organic content of the soil.

It increases the inorganic content of the soil.

It kills micro organisms in the soil.

Environmental impact of tillage

It aerates the soil

It leads to leaching.

It leads to loss of fertility.

Environmental impact of different farming methods

Shifting cultivation results to loss of vegetation

Monocropping leads to loss of soil fertility and rapid spread of pests and diseases in the farms.

Overcrowding

Lack of space leads to overcrowding. Causes of overcrowding are: barriers, immigration and increase in birthrate.

Environmental impact of overcrowding:

Epidemics, fighting, cannibalism, decrease in natality.

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General Impact of Environmental Degradation

Human activity is causing environmental degradation, which is the deterioration of the environment through depletion of resources such as air, water and soil; the destruction of ecosystems; habitat destruction; the extinction of wildlife; and pollution. It is defined as any change or disturbance to the environment perceived to be deleterious or undesirable.

Mass extinction, defaunation, decline in biodiversity, Holocene extinction, biodiversity loss and Ecological collapse

Biodiversity generally refers to the variety and variability of life on Earth, and is represented by the number of different species there are on the planet. Since its introduction, Homo sapiens (the human species) has been killing off entire species either directly (such as through hunting) or indirectly (such as by destroying habitats), causing the extinction of species at an alarming rate. Humans are the cause of the current mass extinction, called the Holocene extinction, driving extinctions to 100 to 1000 times the normal background rate. Though most experts agree that human beings have accelerated the rate of species extinction, some scholars have postulated without humans, the biodiversity of the Earth would grow at an exponential rate rather than decline. The Holocene extinction continues, with meat consumption, overfishing, ocean acidification and the amphibian crisis being a few broader examples of an almost universal, cosmopolitan decline in biodiversity. Human overpopulation (and continued population growth) along with profligate consumption are considered to be the primary drivers of this rapid decline. A 2017 statement by 15,364 scientists from 184 countries warned that, among other things, this sixth extinction event unleashed by humanity could annihilate many current life forms and consign them to extinction by the end of this century.

Defaunation is the loss of animals from ecological communities.

It is estimated that more than 50 percent of all wildlife has been lost in the last 40 years. It is estimated that by 2022, 70% of the world's wildlife will be lost. In South America, this is believed to be a 70 percent loss. A May 2018 study published in PNAS found that 83% of wild mammals, 80% of marine mammals, 50% of plants and 15% of fish have been lost since the dawn of human civilization. Currently, livestock make up 60% of the biomass of all mammals on earth, followed by humans (36%) and wild mammals (4%).

Death of coral reefs

Because of human overpopulation, coral reefs are dying around the world. In particular, coral mining, pollution (organic and non-organic), overfishing, blast fishing and the digging of canals and access into islands and bays are serious threats to these ecosystems. Coral reefs also face high dangers from pollution, diseases, destructive fishing practices and warming oceans. In order to find answers for these problems, researchers study the various factors that impact reefs. The list of factors is long, including the ocean's role as a carbon dioxide sink, atmospheric changes,

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ultraviolet light, ocean acidification, biological virus, impacts of dust storms carrying agents to far flung reefs, pollutants, algal blooms and others. Reefs are threatened well beyond coastal areas.

General estimates show approximately 10% world's coral reefs are already dead It is estimated that about 60% of the world's reefs are at risk due to destructive, human-related activities. The threat to the health of reefs is particularly strong in Southeast Asia, where 80% of reefs are endangered.

Global Warming

Global warming is the result of increasing atmospheric carbon dioxide concentrations which is caused primarily by the combustion of fossil energy sources such as petroleum, coal, and natural gas, and to an unknown extent by destruction of forests, increased methane, volcanic activity and cement production. Such massive alteration of the global carbon cycle has only been possible because of the availability and deployment of advanced technologies, ranging in application from fossil fuel exploration, extraction, distribution, refining, and combustion in power plants and automobile engines and advanced farming practices. Livestock contributes to climate change both through the production of greenhouse gases and through destruction of carbon sinks such as rain-forests. According to the 2006 United Nations/FAO report, 18% of all greenhouse gas emissions found in the atmosphere are due to livestock. The raising of livestock and the land needed to feed them has resulted in the destruction millions of acres of Rainforest and as global demand for meat rises, so too will the demand for land. Ninety-one percent of all rainforest land deforested since 1970 is now used for livestock Potential negative environmental impacts caused by increasing atmospheric carbon dioxide concentrations are rising global air temperatures, altered hydrogeological cycles resulting in more frequent and severe droughts, storms, and floods, as well as sea level rise and ecosystem disruption.

Ozone Layer Depletion

The ozone layer is basically found at a height of about 20-30km above sea level. The ozone layer provides a protective layer, which prevents the penetration of the sun's harmful ultra-violet rays. Dr Joe Farmer\s finding in 1985 has proved that the ozone layer depletion was even greater than predicted. The pollution of the atmosphere from the release of Chlorofluoro Carbon (CFC) gases causes the depletion of the ozone layer and this result in environmental degradation. Most of the skin diseases and low productivity in agriculture are caused by ultra-violet rays. Also an increase in ultra-violet rays affects water bodies, disturbs aquatic life which supports the food chain, and causes the death of fishes that feed us..It also causes the deterioration of synthetic materials such as paints and other products used in the building industry, invariably causing deterioration of housing delivery to our ever-increasing human population.

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Land degradation

Land degradation is a process in which the value of the biophysical environment is affected by a combination of human-induced processes acting upon the land. It is viewed as any change or disturbance to the land perceived to be deleterious or undesirable. Natural hazards are excluded as a cause; however human activities can indirectly affect phenomena such as floods and bush fires.

This is considered to be an important topic of the 21st century due to the implications land degradation has upon agronomic productivity, the environment, and its effects on food security. It is estimated that up to 40% of the world's agricultural land is seriously degraded. Others include:

Desertification

Ocean acidification

Ozone depletion

Water degradation

Disruption of the nitrogen cycle

Human impact on the nitrogen cycle

Of particular concern is N2O, which has an average atmospheric lifetime of 114–120 years, and is 300 times more effective than CO2 as a greenhouse gas. NOx produced by industrial processes, automobiles and agricultural fertilization and NH3 emitted from soils (i.e., as an additional byproduct of nitrification) and livestock operations are transported to downwind ecosystems, influencing N cycling and nutrient losses. Six major effects of NOx and NH3 emissions have been identified:

- decreased atmospheric visibility due to ammonium aerosols (fine particulate matter [PM])
- elevated ozone concentrations
- ozone and PM affects human health (e.g. respiratory diseases, cancer)
- increases in radiative forcing and global warming
- decreased agricultural productivity due to ozone deposition
- ecosystem acidification and eutrophication.

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POLLUTION

Pollution is the introduction of contaminants into the natural environment that causes adverse change. Pollution can take the form of chemical substances or energy, such as noise, heat or light. Pollutants, the components of pollution, can be either foreign substances/energies or naturally occurring contaminants. Pollution is often classed as point source or nonpoint source pollution. In 2015, pollution killed 9 million people in the world.

The major forms of pollution are listed below along with the particular contaminant relevant to each of them:

- Air pollution: the release of chemicals and particulates into the atmosphere. Common gaseous pollutants include carbon monoxide, sulfur dioxide, chlorofluorocarbons (CFCs) and nitrogen oxides produced by industry and motor vehicles. Photochemical ozone and smog are created as nitrogen oxides and hydrocarbons react to sunlight. Particulatmatter, or fine dust is characterized by their micrometre size PM_{10} to $PM_{2.5}$.
- Light pollution: includes light trespass, over-illumination and astronomical interference.
- Littering: the criminal throwing of inappropriate man-made objects, unremoved, onto public and private properties.
- Noise pollution: which encompasses roadway-noise, aircraft noise, industrial noise as well as high-intensity sonar.
- Plastic pollution: involves the accumulation of plastic products and microplastics in the environment that adversely affects wildlife, wildlife habitat, or humans.
- Soil contamination occurs when chemicals are released by spill or underground leakage. Among the most significant soil contaminants are hydrocarbons, heavy metals. MTBEherbicides, pesticides and chlorinated hydrocarbons.
- Radioactive contamination, resulting from 20th century activities in atomic physics, such as nuclear power generation and nuclear weapons research, manufacture and deployment. (See alpha emitters and actinides in the environment.)
- Thermal pollutio, is a temperature change in natural water bodies caused by human influence, such as use of water as coolant in a power plant.
- Visual pollution, which can refer to the presence of overhead power lines, motorway billboards, scarred landforms (as from strip mining), open storage of trash, municipal solid waste or space debris.
- Water pollution, by the discharge of wastewater from commercial and industrial waste (intentionally or through spills) into surface waters; discharges of untreated domestic sewage, and chemical contaminants, such as chlorine, from treated sewage; release of waste and contaminants into surface runoff flowing to surface waters (including urban runoff and agricultural runoff, which may contain chemical fertilizers and pesticides; also including human feces from open defecation - still a major problem in many developing countries); groundwater pollution from waste disposal and leaching into the ground, including from pit latrines and septic tanks; eutrophication and littering.

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Pollutants

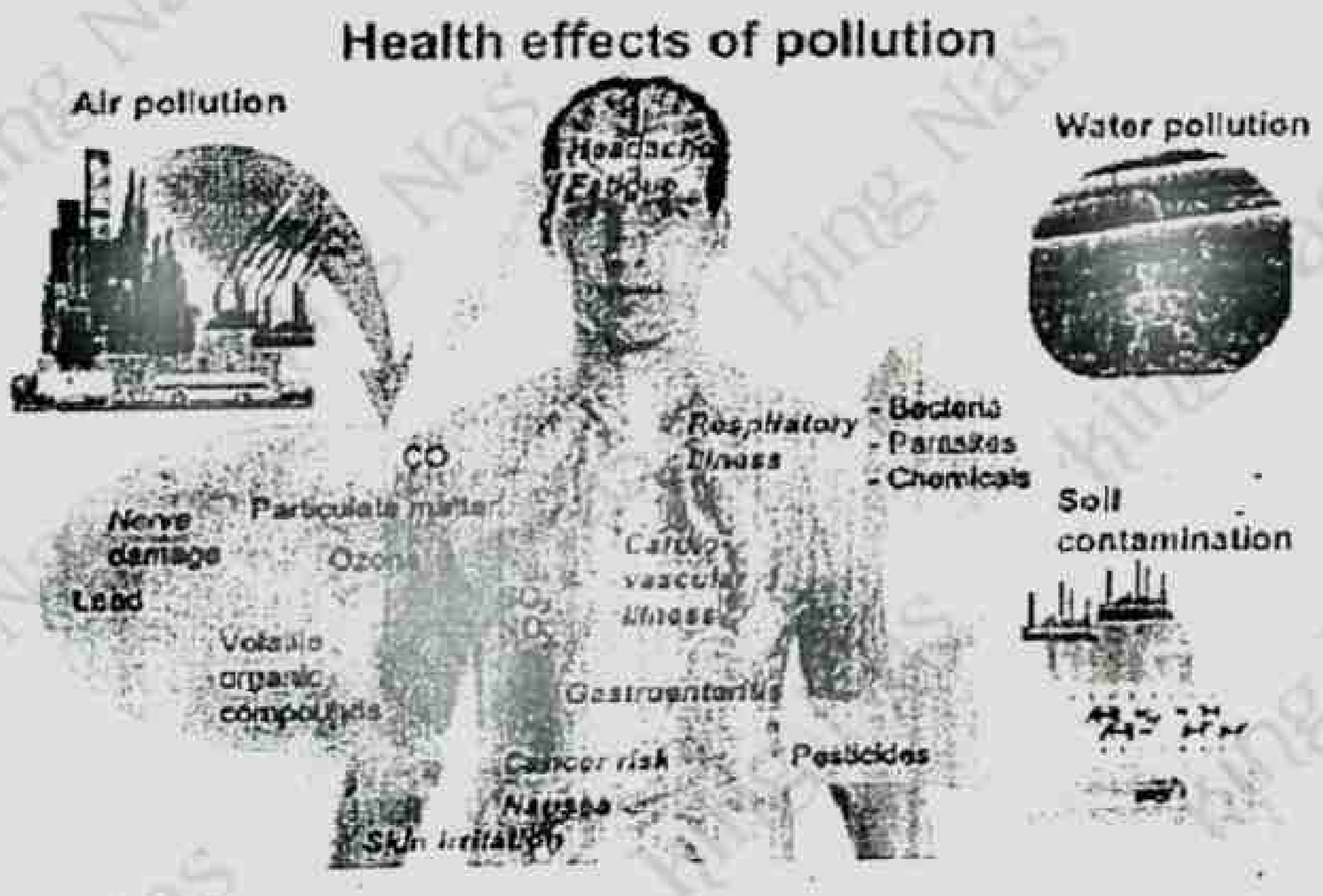
A pollutant is a waste material that pollutes air, water, or soil. Three factors determine the severity of a pollutant: its chemical nature, the concentration and the persistence.

Air pollution comes from both natural and manmade sources. Though globally man made pollutants from combustion, construction, mining, agriculture and warfare are increasingly significant in the air pollution equation.

Motor vehicle emissions are one of the leading causes of air pollution. China, United States, Russia, Mexico, and Japan are the world leaders in air pollution emissions. Principal stationary pollution sources include chemical plants, coal-fired power plants, oil refineries, petrochemical plants, nuclear waste disposal activity, incinerators, large livestock farms (dairy cows, pigs, poultry, etc.), PVC factories, metals production factories, plastics factories, and other heavy industry. Agricultural air pollution comes from contemporary practices which include clear felling and burning of natural vegetation as well as spraying of pesticides and herbicides

About 400 million metric tons of hazardous wastes are generated each year. The United States alone produces about 250 million metric tons. Americans constitute less than 5% of the world's population, but produce roughly 25% of the world's CO2, and generate approximately 30% of world's waste. In 2007, China has overtaken the United States as the world's biggest producer of CO2.

Human Health Effects:



Adverse air quality can kill many organisms including humans. Ozone pollution can cause respiratory disease, cardiovascular disease, throat inflammation, chest pain, and congestion. Water pollution causes approximately 14,000 deaths per day, mostly due to contamination of drinking water by untreated sewage in developing countries. An estimated 700 million Indians have no access to a proper toilet, and 1,000 Indian children die of diarrhoeal sickness every day. Nearly 500 million Chinese lack access to safe

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drinking water. 656,000 people die prematurely each year in China because of air pollution. In India, air pollution is believed to cause 527,700 fatalities a year. Studies have estimated that the number of people killed annually in the US could be over 50,000. Oil spills can cause skin irritations and rashes. Noise pollution induces hearing loss, high blood pressure, stress, and sleep disturbance. Mercury has been linked to developmental deficits in children and neurologic symptoms. Older people are majorly exposed to diseases induced by air pollution. Those with heart or lung disorders are under additional risk. Children and infants are also at serious risk. Lead and other heavy metals have been shown to cause neurological problems. Chemical and radioactive substances can cause cancer and as well as birth defects.

Effects in the Environment

Pollution has been found to be present widely in the environment. There are a number of effects of this:

- Biomagnification describes situations where toxins (such as heavy metals) may pass through trophic levels, becoming exponentially more concentrated in the process.
- Carbon dioxide emissions cause ocean acidification, the ongoing decrease in the pH of the Earth's oceans as CO_2 becomes dissolved.
- wrtThe emission of greenhouse gases leads to global warming which affects ecosystems in many ways.
- Invasive species can out-compete native species and reduce biodiversity. Invasive plants can contribute debris and biomolecules (allelopathy) that can alter soil and chemical compositions of an environment, often reducing native species competitiveness.
- Nitrogen oxides are removed from the air by rain and fertilise land which can change the species composition of ecosystems.
- Smog and haze can reduce the amount of sunlight received by plants to carry out photosynthesis and leads to the production of tropospheric ozone which damages plants.
- Soil can become infertile and unsuitable for plants. This will affect other organisms in the food web.
- Sulfur dioxide and nitrogen oxides can cause acid rain which lowers the pH value of soil.
- Organic pollution of watercourses can deplete oxygen levels and reduce species diversity.

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Regulation and monitoring of pollution

To protect the environment from the adverse effects of pollution, many nations worldwide have enacted legislation to regulate various types of pollution as well as to mitigate the adverse effects of pollution.

Pollution control

Pollution control is a term used in environmental management. It means the control of emissions and effluents into air, water or soil. Without pollution control, the waste products from overconsumption, heating, agriculture, mining, manufacturing, transportation and other human activities, whether they accumulate or disperse, will degrade the environment. In the hierarchy of controls, pollution prevention and waste minimization are more desirable than pollution control. In the field of land development, low impact development is a similar technique for the prevention of urban runoff.

Practices

- Recycling
- Reusing
- Waste minimisation
- Mitigating
- Preventing
- Compost

Pollution control devices for Air pollution control

Thermal oxidizer

Dust collection systems

Baghouses

Cyclones

Electrostatic precipitators

Scrubbers

Baffle spray scrubber

Cyclonic spray scrubber

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    - Ejector venturi scrubber
    - Mechanically aided scrubber
    - Spray tower
    - Wet scrubber
- Sewage treatment
    - Sedimentation (Primary treatment)
    - Activated sludge biotreaters (Secondary treatment; also used for industrial wastewater)
    - Aerated lagoons
    - Constructed wetlands (also used for urban runoff)
- Industrial wastewater treatment
    - API oil-water separators
    - Biofilters
    - Dissolved air flotation (DAF)
    - Powdered activated carbon treatment
    - Ultrafiltration
- Vapor recovery systems
- Phytoremediation

Health Effects of Toxic Chemicals

There are many materials used in the workplace that can be hazardous. However, in order for them to affect your health, they must come into contact with the body or be absorbed into the body. When assessing the potential health effects from working with a particular material, it is necessary to understand the difference between "toxicity" and "hazard".

1. **Toxicity** is the ability of a substance to produce an unwanted effect when the chemical has reached a sufficient concentration at a certain site in the body.

The more toxic a material is, the smaller the amount of it is necessary to be absorbed before harmful effects are caused. The lower the toxicity, the greater the quantity of it is necessary to be absorbed. Generally, the toxicity of a chemical is determined through experiments on animals (quite often rats), and measured in terms of the amount of material necessary to cause death in 50% of the test animals. These values are called LD50 (lethal dose) or LC50 (lethal concentration), and are usually given in weight of material per kg of body weight or airborne concentration of material per set time period respectively.

1. **Hazard** is the probability that this concentration in the body will occur.

Toxicity is an inherent property of the material. A material may be very toxic, but not hazardous, if it is handled properly and is not absorbed into the body. On the other hand, a material may have a very low toxicity, but be very hazardous.

Example:

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- An open container of an acid is much more hazardous than a closed container of the same material.
- Two liquids may possess the same degree of toxicity but present different degrees of hazard:
- One material may be non-irritating to the eyes and nose and odourless. The other may be irritating to the eyes or respiratory system and possess a pungent odour.
- The latter material, because of its warning properties presents a lesser degree of hazard.

In order for toxicants to affect the human system either they must cause damage to external tissues, such as the skin or eyes, or they must be able to enter the body by some mechanism.

Routes of Entry

There are three primary routes of entry into the body: ingestion, skin or eye absorption, inhalation and injection.

Ingestion- This means taking a material into the body by mouth (swallowing). Ingestion of toxic materials may occur as a result of eating in a contaminated work area.

Absorption- Substances that contact the eye and the skin may be either absorbed into the body or cause local effects. For the majority of organic compounds, the contribution from skin absorption to the total exposure should not be neglected.

Inhalation- This means taking a material into the body by breathing it in. In the lungs, very tiny blood vessels are in constant contact with the air we breathe in. As a result, airborne contaminants can be easily absorbed through this tissue. In the occupational environment, this is generally the most important route of entry.

Injection- This means taking in biological or chemical substances into the body when the skin is accidentally punctured with a sharp object (e.g. contaminated needle).

Health Effects – Chronic vs Acute

Once a toxic substance has contacted the body it may have either acute (immediate) or chronic (long term) effects.

Example: Spilling acid on your hand will cause **immediate harm,** i.e. a burn to the skin.

Exposure to asbestos or tobacco smoke may result in lung cancer after as much as twenty years (this is a long term effect).

Exposure – Chronic vs Acute

Exposure can be classified as chronic or acute. In chronic exposures, the dose is delivered at some frequency (daily or weekly usually) over a period of time. In acute exposures, the dose is

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delivered in a single event and absorption is rapid. Usually, a chronic exposure occurs at low concentration and acute exposure at high concentration.

Some materials may only cause harm if given acutely, not having any effect in the long term. Other materials may not exhibit an effect in the short term, but may cause problems after prolonged exposure.

Physiological Classification of Materials

This classification identifies toxic materials on the basis of biologic action.

Irritants – refers to some sort of aggravation of whatever tissue the material comes in contact with. e.g. ammonia, nitrogen dioxide.

Asphyxiants – exert their effects through a depletion of oxygen to the tissues e.g. simple asphyxiants – carbon dioxide, nitrogen, methane, hydrogen chemical asphyxiants – carbon monoxide, hydrogen cyanide, hydrogen sulphide.

Narcotics or Anaesthetics – the main toxic action is the depressant effect upon the Central Nervous System. e.g. – many organics, chloroform, xylene.

Systemic Poisons – the main toxic action includes the production of internal damage e.g. Hepatotoxic agents – toxic effects produce liver damage. eg. carbon tetrachloride.

e.g. Nephrotoxic agents – toxic effects produce kidney damage eg. some halogenated hydrocarbons

Carcinogens – agents/compounds that will induce cancer in humans. e.g. benzene, arsenic, inorganic salts of chromium, nickel, beryllium.

Mutagens – agents that affect the cells of the exposed people in such a way that it may cause cancer in the exposed individual or an undesirable mutation to occur in some later generation. e.g. radiation, variety of chemical agents that alter the genetic message.

Teratogens – Agents or compounds that a pregnant woman takes into her body that generate dy defects in the fetus e.g. Thalidomide, possibly steroids

Sensitizers – Agents that may cause allergic or allergic-like responses to occur. After an initial exposure to a substance an individual may become sensitized to that substance. Subsequent exposures to the same substance, often at a much lower concentration than before, produce an allergic response. This response may be a skin rash (dermatitis) or an asthmatic-like attack, depending on the route of exposure. e.g. cutting oils, isocyanates in polyurethane foam operations and paint spraying operations, some laboratory solvents.

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Toxic chemicals present in the environment: sources and toxic action

- There are a large number of chemicals in the environment. Some of these are toxic and the rest are non-toxic. Many metals which are listed as environmental hazards are essential dietry trace elements required for normal growth and development of living things. These elements include Al, Sb, As, Be, Bi, Cd, Co, Cu, In,Pb, Mo, Ag, Te, Sn, Ti, W, U, and Zc. In such cases, it is the amount or the concentration of the respective element which makes it toxic. Even the well known toxic element like As, Pb, and Cd are required in trace quantities for the growth of animals. A typical example is that of Al, which biologically inert, but it causes brain damage, bone disease and anaemia in patients subjected to haemodialysis using water containing 100-1000 part per billion of aluminum. In fact 100-1000 ppb is the normal concentration of aluminium in drinking water where aum is used for water treatment.

The fact is that everyone is exposed to toxic chemicals at home, at work, at school and in the outdoor environment. The workplace is an important source of exposure. Thus, roofers, construction workers and automobile engineers are exposed to particulate matter besides other pollutants. Farm workers are exposed to pesticides. Anyone who cleans anything may be exposed to chlorinated solvents. Office going people come in contact with photocopier, tonners, correction fluids and rubber, cement. Hospital workers surrounded by disinfectants. Everyone is exposed at home to cleaners' detergents, paints and others. Animals, plants and microorganisms encounter many of these substances as environmental pollutants. It is in fact because of the presence of toxic chemicals in the environment that environmental toxicology is important.

Toxic Chemicals in the Air

The usual major pollutants (like oxides of carbon, nitrogen and sulphur) particulate pollutants and metallic pollutants pose serious health hazards, it is necessary to control these pollutants. Besides the above, the US Environmental Protection Agency along with consumer products Safety Commission has identified extremely hazardous substances in the atmosphere. These are acrylonitrate, arsenic asbestors, berellium, Cadmium, chlorinated solvents, chlorofluorocarbon, chromates, coke-oven emission, diethylstilbestrol, dibromochloropropene, ethylbromide, ethylene oxide, lead, mercury, nitrosoamines, ozone, polybrominated biphenyls, polychlorinated biphenyls, radiation, sulphur dioxide, vinyl chloride, and toxic waste disposal emissions and leachates.

Toxic Elements in Water

A list of toxic elements found in natural and waste water is given in Table 1 below. Some of these are essential at low levels (necessary as nutrients for animal and plant life) but are toxic at higher levels. Besides the toxic elements the water bodies contain a large number of pesticides, which are primarily agricultural runoffs.

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Element	Sources	Effects and Significance
Arsenic	Mining by-products, pesticides, chemical waste	Toxic, possibly carcinogenic
Cadmium	Industrial discharge, mining waste, metal plating, water pipes	Replaces zinc biochemically, causes high blood pressure, kidney damage, destruction of testicular tissue and red blood cells.
Beryllium	Coal, nuclear power and space, industries.	Acute and chronic toxicity, possibly carcinogenic
Boron	Coal, detergent formulations industrial wastes.	Toxic to some plants
Chromium	Metal plaiting, cooling tower water additives, (chromate) normally found as Cr(vi) in polluted water.	Essential trace elements possibly carcinogen as Cr(vi).
Copper	Metal plating, industrial and domestic wastes, mining, mineral leaching	Essentially trace elements not very toxic to animals, toxic to plants and algae at moderate level.
Fluorine (fluoride ion)	Natural geological sources, industrial wastes water additives	Prevents both decay at about 1mg/l, causes mottled teeth and bone damage at about 5mg/l.
Lead	Industry, mining, plumbing, coal, gasoline	Toxic, (anaemia, kidney disease, nervous disorder), wildlife destroyed.
Manganese	Mining, industrial waste, acid mine drainage, microbial action on manganese at low pH	Relatively non-toxic to animals, toxic to plants at higher levels, stains materials (bathroom fixtures and clothing).
Mercury	Industrial waste, mining, natural sources.	Toxic to animals.
Selenium	Natural geological sources, sulphur, coal.	Essential at low level but toxic at higher level.
Zinc	Industrial waste, metal plating, plumbing.	Essential in metallo-enzymes, toxic to plants at higher levels.

Toxic Waste in Soils

Soil has been used as a dumping ground for most of the waste products including toxic wastes. The main sources of these wastes are industrial wastes, urban and domestic wastes, radioactive wastes, agrochemicals like fertilizers and pesticides, plastics and polymers, chemical and metallic pollutants, biological agents.

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CONSERVATION OF NATURAL RESOURCES

Conservation is the planned use or exploitation of natural resources to ensure their continuous use and preserve the original nature of the environment. Natural resources are divided into two they are renewable and non-renewable natural resources.

Renewable natural resources are the resources that can be replaced. They are water, air, food, forest, soil; animals etc. non-renewable are the ones that cannot be replaced in future. The quantity of non-renewable resources is fixed on earth. Examples of non-renewable natural resources are crude oil, cola, bauxite limestone, tin, oil, gas etc.

METHODS OF CONSERVATION

Methods of conserving water

Irrigation processes, construction of dams and reservoirs. control of the use of underground water effective control of water pollution, treatment of and recycling of used water and proper channeling of rain water to wells, tanks and collecting centers.

Benefits of water conservation

Source of food, source of water for irrigation, domestic use, hydro-electricity, transportation.

Water preserves natural resources

Methods of conserving forest

Establishment of a forest reserve, legislation, restricting the felling down of economic trees, proper recycling of paper and other forest products, proper treatment of the soil, reforestation, and efficient forest management measures.

Benefits of forest conservation

Supplies timber and other forest products

Checks erosion and desert encroachment

Preserves natural species

Encourages rainfall and helps to maintain natural succession

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Methods of conserving wildlife

By restricting hunting processes, legislation to maintain population of animals, artificial stocking and game farming

Benefits of conserving wildlife

Source of food/protein and raw material for industries example hides and skins, ivory

Boosts tourism education or recreation

Can be used for ecological or scientific studies.

Methods of conserving fisheries

Control of water pollution, artificial stocking, fishing regulations, construction of fish ponds, and seed multiplication, control of aquatic weeds.

Benefits of conserving fisheries

- Source of food/protein

Boosts tourism education or recreation

Can be used for ecological or scientific studies.

Methods of conserving soil

Through the control of soil erosion, improvement of soil fertility through the addition of manures, planting of cover crops, crop rotation, nitrogen cycle etc. By afforestation to prevent wind erosion and desertification; through the control of soil erosion.

Benefits of conserving soil

Source of plant nutrients for growth

Helps to preserve soil fertility.

It is a reservoir of water for soil organisms

A source of raw materials for some industries e.g use of clay in ceramics industries

Methods of conserving mineral resources

By substituting fossil fuel with solar energy

Recycling used mineral products

Preventing accidental waste of minerals example oil spillage

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Efficient use of metal scraps example iron and tin.

Reduction in the use of mineral resources generally.

Benefits of conserving mineral resources

Mineral resources are source of energy, raw materials for industries and source for common salt for food, and source of foreign exchange.

Benefits of petroleum and natural gas

Ensures the continuous supply of raw materials for petrochemical industries

Continuous use of fossil fuel

Ensures employment opportunities.

Problems encountered in conservation

Economic problems: Lack of money to carry out the conservation processes.

Limited land area for development farming: This has led to the destruction of natural forests.

Lmited source of energy: this also has led to destruction of forests for firewood

Maintenance of river for hydro-electricity and irrigation has resulted in the alteration of the ecosystem.

Overgrazing, overfishing or over cropping activities.

The nuclear power which is an alternative source of energy is dangerous and causes thermal pollution.

Bush burning bush fire.

Lack of public awareness and good management.

AGENCIES RESPONSIBLE FOR CONSERVATION

River Basin Authorities

PHCN ((Power Holding Company of Nigeria).

Ministry of Agriculture: 1. Department of Agriculture 2. Department of wildlife conservation 3. Department of Fisheries

Ministry of Energy Resources.

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National Environmental Sanitation Authority

Nigerian Conservation Foundation

Urbanization: Causes and Effects

More than half of the world's population lives in urban areas. Due to the ongoing urbanization and growth of the world's population, there will be about 2.5 billion more people added to the urban population by 2050, mainly in Africa and Asia. The world's urban areas are highly varied, but many cities and towns are facing problems such as a lack of jobs, homelessness and expanding squatter settlements, inadequate services and infrastructure, poor health and educational services and high levels of pollution.

Urbanization is an increase in the number of people living in towns and cities. Urbanization occurs mainly because people move from rural areas to urban areas and it results in growth in the size of the urban population and the extent of urban areas. These changes in population lead to other changes in land use, economic activity and culture. Historically, urbanization has been associated with significant economic and social transformations. For example, urban living is linked with higher levels of literacy and education, better health, lower fertility and a longer life expectancy, greater access to social services and enhanced opportunities for cultural and political participation. However, urbanization also has disadvantages caused by rapid and unplanned urban growth resulting in poor infrastructures such as inadequate housing, water and sanitation, transport and health care services.

The rapid increase in urban populations has meant that peri-urban areas are growing much more quickly than formal urban centers. **Peri-urban** areas are those areas immediately around a town or city. They are areas in transition from countryside to city (rural to urban), often with undeveloped infrastructure, where health and sanitation services are under pressure and where the natural environment is at risk of degradation.

Causes of urbanization

Urbanization in the developing world occurs for two main reasons: the natural increase of population and rural to urban migration.

Natural increase of population

As birth rates decline over time, according to the demographic transition model, the role of natural increase in determining the pace of urban population growth becomes less important in comparison to migration.

Rural to urban migration

In developing countries, urbanization usually occurs when people move from villages to settle in cities in hope of gaining a better standard of living. The movement of people from one place to another is called **migration**. Migration is influenced by economic growth and development and

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by technological change and possibly also by conflict and social disruption. It is driven by **pull factors** that attract people to urban areas and **push factors** that drive people away from the countryside.

Employment opportunities in cities are one of the main pull factors. Many industries are located in cities and offer opportunities of high urban wages. There are also more educational institutions providing courses and training in a wide range of subjects and skills. People are attracted to an urban lifestyle and the 'bright lights' of city life. All of these factors result in both temporary and permanent migration to urban areas.

Poor living conditions and the lack of opportunities for paid employment in rural areas are push factors. People are moving away from rural areas because of poor health care and limited educational and economic opportunities as well as environmental changes, droughts, floods, lack of availability of sufficiently productive land, and other pressures on rural livelihoods.

Rural to urban migration can be a selective process, as some types of people are more likely to move than others. One of the factors involved is gender, because employment opportunities vary greatly with different jobs for men and women. Another factor is age. Young people are more likely to move to towns, with more elderly people and children left in rural areas. Selectivity in migration affects the population in both the rural and the urban areas. If more men move to towns and cities than women, this leaves a predominantly female society in rural areas.

Impacts of urbanization

Although people are pulled towards the advantages of cities, the impacts of urbanization are mixed. First we will look at the many positive impacts of urbanization before going on to describe some of the challenges created by rapid unplanned urban growth.

Thriving towns and cities are an essential element of a prosperous national economy. The gathering of economic and human resources in one place stimulates innovation and development in business, science, technology and industry. Access to education, health, social services and cultural activities is more readily available to people in cities than in villages. In cities, child survival rates are better than in rural areas because of better access to health care. The density of urban populations makes it easier and less costly for the government and utilities to provide essential goods and services. For example, the supply of basic facilities such as fresh water and electricity can be achieved with less effort and less cost per person.

Schools, colleges and universities are established in cities to develop human resources. A variety of educational courses are available, offering students a wide choice for their future careers. People of many classes and religions live and work together in cities, which creates better understanding and harmony and helps break down social and cultural barriers. Cities also have advanced communication and transport networks.

However, these many benefits of urban life do not apply to all. Rapid population increases and unplanned growth create an urban sprawl with negative economic, social, and environmental consequences. In Nigeria, the rate of urban growth often strains the capacity of local and national

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government to provide urban residents with even the most basic services of housing, water supply, sewerage and solid waste disposal.

Housing

In developing countries, about a third of urban inhabitants live in impoverished slums and squatter settlements. **Slums** are urban areas that are heavily populated and have sub-standard housing with very poor living conditions, creating several problems.

In Nigeria, about 60% of the houses in the city were classed as slums due to the physical deterioration of its housing, overcrowding, high density, poor access and lack of infrastructure services.

Slum areas typically suffer from:

- poor housing with small, overcrowded houses built very close together using inadequate materials and with uncertain electricity supply
- restricted access to water supplies
- little or no sanitation/latrine facilities and no solid waste disposal, which leads to a polluted and degraded local environment
- inadequate health care facilities which, coupled with the poor living conditions, increases sickness and death rates
- insecure living conditions – slum dwellers may be forcibly removed by landowners or other authorities.

Water supply and sanitation

The provision of water and sanitation services to growing urban settlements, peri-urban and slum areas presents critical challenges. The increased demand for water from the growing population can place added stress on already stretched resources. In and around cities, water is commonly in short supply and subject to increasing competition by different users. Urban growth leads to increasing demand for water for industrial and domestic use, which conflicts with agricultural demands.

It is especially difficult to provide water and sanitation services to deprived areas and the poorest people. Many people in these areas live without access to safe drinking water and proper sanitation. Even where adequate water supplies are available, sanitation and wastewater disposal are often inadequate or missing. Pit latrines and septic tanks are the usual methods for human waste disposal but they have limited capacity and are not always adequate to cope with the quantity of waste produced by many people living close together. Overflowing latrines and septic tanks contaminate surface water and create a serious health risk.

The lack of these essential services threatens not only the health and the environment of people in slum areas, but also that of people living in formal urban areas. This is true of many cities with a million or more inhabitants, as well as smaller cities and towns.

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Wastes and pollution

Urbanisation affects land, water, air and wildlife because of the number of people, the amount of buildings and construction, and the increased demands on resources. It has impacts on the physical environment in several ways.

Water quality

In developing countries, including Nigeria, many rivers in urban areas are more like open sewers. The lack of sanitation and sewerage systems has a dramatic impact on urban watercourses. People use the rivers to dispose of all their wastes from homes, industries and commercial businesses. Wastewater from human settlements contains organic material and nutrients; industrial wastewater contains many different types of toxic pollutant. These make the water unsafe for humans to use for many purposes including drinking and irrigation, as well as harming the fish and other animals and plants living in the water. Any change to the quality of surface water also affects groundwater because they are linked by the processes of the water cycle so pollutants from the surface will infiltrate down and contaminate soil and groundwater as well.

Solid waste

In many towns and cities solid waste management is inefficient or non-existent. Solid waste management means the proper collection, transfer, recycling and disposal of all the solid material we throw away, including plastics, paper and cardboard, food wastes, electrical waste, etc. It also includes industrial, hospital and institutional wastes which often contain pathogens as well as hazardous and toxic chemicals, which need special care.

Urban waste often ends up in illegal dumps on streets, open spaces, wastelands, drains or rivers. This is frequently a problem in peri-urban areas, which are convenient for dumping wastes because of the availability of open space and ease of access from central urban areas. This can lead to the pollution of groundwater and surface waters which may be used as a source for drinking water. Sometimes the wastes are collected and taken to legalised waste disposal sites but these are not always properly managed to protect water bodies and groundwater.

The combustion of solid waste creates yet another environmental problem. People want to get rid of the wastes and they will burn them in their backyards if there is no collection system.

Air quality

Air quality in towns and cities is frequently very poor as a result of air pollution from many different sources. These include:

- vehicle exhausts
- smoke from domestic fires

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- outputs from factory chimneys
- diesel-powered generators
- dust from construction works and city streets.

Poor air quality has a significant impact on the health of many urban residents as well as leaving a damaging and unsightly layer of dust on plants, buildings and other surfaces.

Health

Urbanisation can have both positive and negative effects on health. The main benefits are associated with easier access to hospitals, clinics and health services in general. If you live close to these services you can reach a doctor in minutes rather than hours or days, so this improves emergency care and general health. There are also benefits from easier access to sources of information such as radio and television which may be used to communicate information about health to the general public. For example, women living in towns and cities are more likely to be informed about family planning, which results in reduction in family size and less frequent childbirth, with consequent benefits to general health.

However, urban life can also damage health. Poor environment, housing and living conditions are the main reasons for poor health in urban areas. Contamination of water sources can cause epidemics of waterborne disease. Close proximity to other people can make the spread of many types of infectious disease more likely. The polluted air can also cause respiratory disease and contribute to premature deaths among more vulnerable sections of the population such as older people and children.

Food

Population movements also put pressure on food supplies and on food distribution. As people migrate to the cities, they tend to use purchased food instead of their own crops and this makes them more vulnerable to changes in food prices. As the population grows and the demand for water and land increases, it becomes difficult to increase food production in a sustainable way. The increase in urban demand, combined with a loss of agricultural land, means more pressure on rural people to produce food for the growing number of urban people.

Furthermore, pollution from urban areas can disrupt food supply. For example, fisheries are often damaged by urban domestic wastes and liquid effluents from city-based industries. Effluent is another word for wastewater that flows out from a source.

Economic and social systems

The process of urbanization has positive as well as negative economic and social changes. The positive effects include economic development, and education. However, urbanisation places stresses on existing social services and infrastructure. Crime, prostitution, drug abuse and street children are all negative effects of urbanisation. Also there tends to be a lack of social support for children in school and home by their hard-working, usually poor, parents. Inadequate income, overcrowded housing and poor living conditions create a fertile ground for the development of violence. Violent crime is more visible in the cities than in rural areas and it affects people's

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everyday life, their movements and the use of public transport. Crime in the city can create a sense of insecurity in its inhabitants. This unsafe feeling in city streets separates residential areas into higher-income and lower-income groups, which reduces the sense of community and forms areas with dissimilar incomes, costs and security levels.

Health Effects of Environmental Degradation

The urban environment is an important factor in determining the quality of life in urban areas. Some urban environmental problems include inadequate water and sanitation, lack of rubbish disposal, and industrial pollution. Unfortunately, reducing the problems and ameliorating their effects on the urban population are expensive.

The health implications of these environmental problems include respiratory infections and other infectious and parasitic diseases.

Capital costs for building improved environmental infrastructure — for example, investments in a cleaner public transportation system such as a subway — and for building more hospitals and clinics are higher in cities, where wages exceed those paid in rural areas.

And urban land prices are much higher because of the competition for space. But not all urban areas have the same kinds of environmental conditions or health problems. Some research suggests that indicators of health problems, such as rates of infant mortality, are higher in cities that are growing rapidly than in those where growth is slower.

Environmental Effects of Urbanization

Urban populations interact with their environment. Urban people change their environment through their consumption of food, energy, water, and land. And in turn, the polluted urban environment affects the health and quality of life of the urban population.

People who live in urban areas have very different consumption patterns than residents in rural areas. For example, urban populations consume much more food, energy, and durable goods than rural populations. In China during the 1970s, the urban populations consumed more than twice as much pork as the rural populations who were raising the pigs.[11] Witheconomic development, the difference in consumption declined as the rural populations ate better diets. But even a decade later, urban populations had 60 percent more pork in their diets than rural populations. The increasing consumption of meat is a sign of growing affluence in Beijing; in India where many urban residents are vegetarians, greater prosperity is seen in higher consumption of milk.

Urban populations not only consume more food, but they also consume more durable goods. In the early 1990s, Chinese households in urban areas were two times more likely to have a TV, eight times more likely to have a washing machine, and 25 times more likely to have a refrigerator than rural households. This increased consumption is a function of urban labor markets, wages, and household structure.

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areas than in rural villages. For example, urban populations have many more cars than rural Energy consumption for electricity, transportation, cooking, and heating is much higher in urban populations per capita. Almost all of the cars in the world in the 1930s were in the United States. Today we have a car for every two people in the United States. If that became the norm, in 2050 there would be 5.3 billion cars in the world, all using energy.

In China the per capita consumption of coal in towns and cities is over three times the consumption in rural areas.[14] Comparisons of changes in world energy consumption per capita and GNP show that the two are positively correlated but may not change at the same rate.[15] As countries move from using noncommercial forms of energy to commercial forms, the relative price of energy increases. Economies, therefore, often become more efficient as they develop because of advances in technology and changes in consumption behavior. The urbanization of the world's populations, however, will increase aggregate energy use, despite efficiencies and new technologies. And the increased consumption of energy is likely to have deleterious environmental effects.

Urban consumption of energy helps create heat islands that can change local weather patterns and weather downwind from the heat islands. The heat island phenomenon is created because cities radiate heat back into the atmosphere at a rate 15 percent to 30 percent less than rural area. The combination of the increased energy consumption and difference in albedo (radiation) means that cities are warmer than rural areas (0.6 to 1.3 C). And these heat islands become traps for atmospheric pollutants. Cloudiness and fog occur with greater frequency. Precipitation is 5 percent to 10 percent higher in cities; thunderstorms and hailstorms are much more frequent, but snow days in cities are less common.

Urbanization also affects the broader regional environments. Regions downwind from large industrial complexes also see increases in the amount of precipitation, air pollution, and the number of days with thunderstorms. Urban areas affect not only the weather patterns, but also the runoff patterns for water. Urban areas generally generate more rain, but they reduce the infiltration of water and lower the water tables. This means that runoff occurs more rapidly with greater peak flows. Flood volumes increase, as do floods and water pollution downstream.

Many of the effects of urban areas on the environment are not necessarily linear. Bigger urban areas do not always create more environmental problems. And small urban areas can cause large problems. Much of what determines the extent of the environmental impacts is how the urban populations behave — their consumption and living patterns — not just how large they are.

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